



Technology Stack

- **Mobile App:** React Native (iOS & Android native), Reanimated 3, MapLibre GL radar
- **Backend / API:** Python 3.11 ([FastAPI](#) Async), Node.js Gateway, WebSockets ([WSS](#)), REST
- **AI / ML & Graph DL:** PyTorch Geometric ([ST-GAT](#) Corridor Attention), ONNX Runtime (<10ms)
- **Physics Kinematics:** Newton-Davis tractive engine, grade, air drag & RK4 ODE numerical solver
- **Data Streaming:** Apache Kafka (100k+ event/s KRaft), Redis 7.2 ([Pub/Sub](#) in-memory cache)
- **Spatial & Time-Series:** PostgreSQL 16 + [PostGIS](#) (1D rail chainage LRS), TimescaleDB
- **Railway Ingestion:** BEL RTIS (NavIC 30s GPS), CRIS FOIS freight, COA logs, e-Caution TSR
- **Security & Hosting:** OAuth 2.0 / JWT, AES-256 GCM, Docker, Kubernetes on RailCloud / NIC

100% ZERO-HARDWARE Taps existing statutory Indian Railways GPS & CRIS feeds. No engine modifications.

Powered By

Python

PyTorch

React Native

Kafka

Redis

PostgreSQL

PostGIS

FastAPI

Docker

Kubernetes

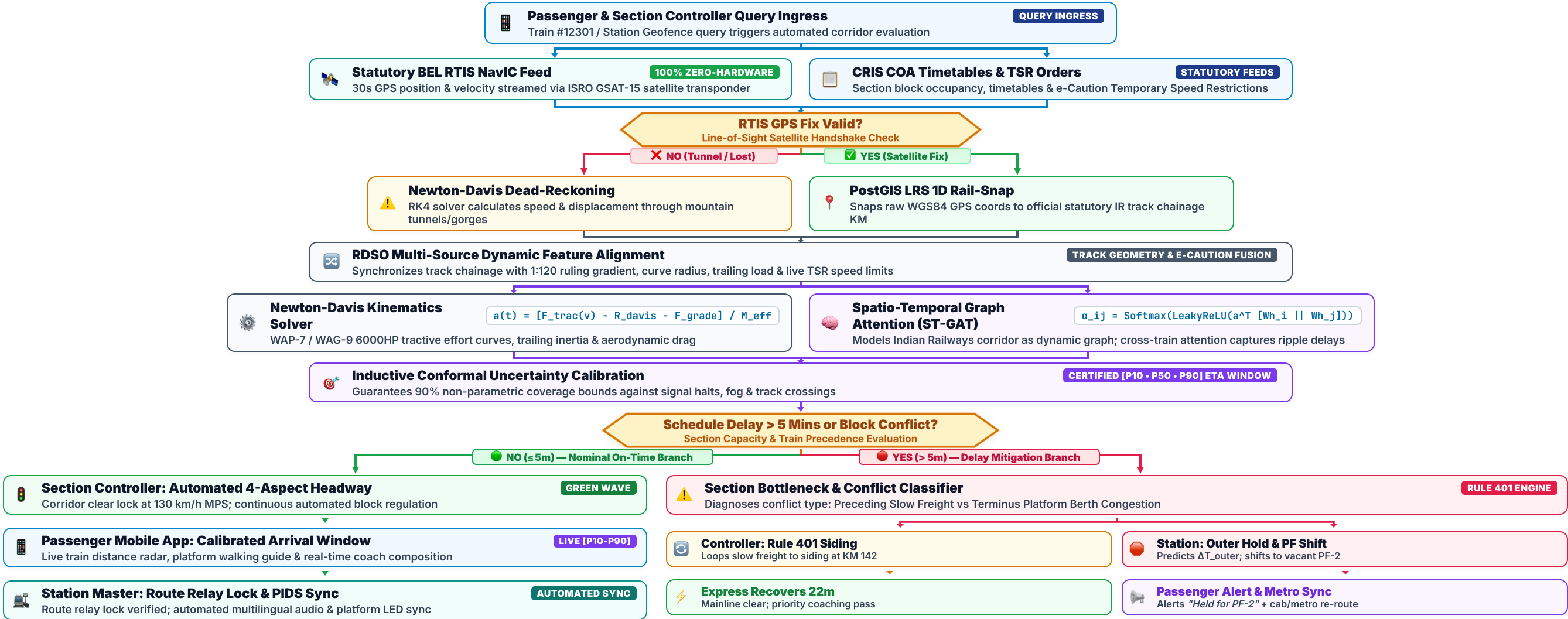
ISRO NavIC

CRIS / IR

TECHNICAL APPROACH

GATI-SETU: Dynamic Closed-Loop Operational Decision Flowchart

Closed-Loop Ground Reality Engine • Statutory 30s NavIC Ingestion



MASTER IMPLEMENTATION ARCHITECTURE (INPUT • PROCESSING • OUTPUT PIPELINE)

End-to-End Operational Lifecycle

